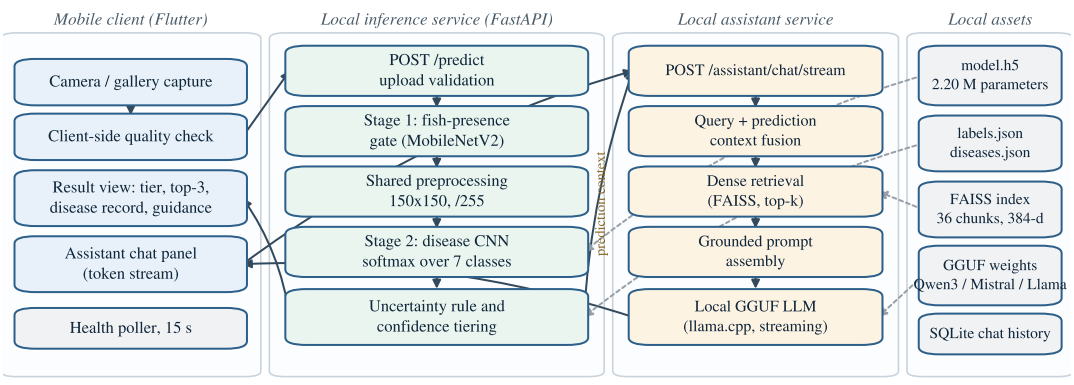
Mobile client (Flutter)
Camera / gallery capture
Client-side quality check
Result view: tier, top-3, disease record, guidance
Assistant chat panel (token stream)
Health poller, 15 s
Local inference service (FastAPI)
POST /predict upload validation
Stage 1: fish-presence gate (MobileNetV2)
Shared preprocessing 150x150, /255
Stage 2: disease CNN softmax over 7 classes
Uncertainty rule and confidence tiering
prediction context
Local assistant service
POST /assistant/chat/stream
Query + prediction context fusion
Dense retrieval (FAISS, top-k)
Grounded prompt assembly
Local GGUF LLM (llama.cpp, streaming)
Local assets
model.h5 2.20 M parameters
labels.json diseases.json
FAISS index 36 chunks, 384-d
GGUF weights Qwen3 / Mistral / Llama
SQLite chat history